

Aditya Kulkarni

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github.com/Aditya-k24 | Raleigh, NC (Open to relocation)

PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) with production full-stack TypeScript/React engineering experience and a strong track record building AI-powered workflow automation tools. At Isomer AI, delivered 10+ backend features and AI agent tooling across 3 LLM providers in production -- 394 commits, 7 deployments, CI/CD with Docker and Sentry. Built Next.js 16 frontends, REST microservices, and event-driven real-time architectures handling 2,000 async requests/second. Experienced in code reviews, agile workflows, production on-call, and integrating AI-assisted development tools into daily engineering practice.

CORE COMPETENCIES

TypeScript / JavaScript

React / Next.js

AI-Powered Editorial Tools

REST API Design

Docker / AWS / CI/CD

Full-Stack Feature Delivery

Real-Time Systems

Agile Development

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern — Chicago, IL (Remote)

- Built 10+ full-stack features including **AI-powered workflow automation** components using TypeScript, Node.js, Express, PostgreSQL, Redis, and Pusher (real-time events) -- 7 production deployments with Docker, GitHub Actions CI/CD, and Sentry observability
- Implemented multi-provider **LLM agent tooling** across Anthropic Claude, OpenAI GPT-4o, and Google Gemini with MCP; built document processing pipelines (PDF, email via Microsoft Graph API, S3); contributed 394 commits and reviewed pull requests across the engineering team

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern — Mumbai, India

- Automated deployments on **AWS and GCP**; integrated Grafana monitoring dashboards and supported model validation and red-teaming for production reliability
- Built 5+ AI-driven workflows including a health claim automation tool using OCR, NLP, and ML; improved claim accuracy by 25%

Capgemini Jun 2023 - Aug 2023

AI/ML Intern — Mumbai, India

- Led end-to-end SDLC for a data-driven system automating support ticket analysis, reducing processing time by 15%; implemented Kafka streaming reducing latency by 20%

NC State University Apr 2025 - Present

Graduate Research Assistant — Raleigh, NC

- Designed a cloud-based data platform serving 100+ researchers worldwide on Google Cloud Run; improved ML model accuracy by 12% via hyperparameter tuning

PROJECTS

CryptoKnight – Trustless AI Trading Agent Next.js 16 + TypeScript

Built a full-stack DeFi application using Next.js 16, TypeScript, viem, and wagmi on the frontend with a Python AI agent backend. Solved the "Delegation Dilemma" via ERC-4337 Account Abstraction -- AI agent trades Polymarket autonomously using scoped Session Keys with on-chain policy enforcement (trade limits, expiry, instant revocation). Built complete MVP at HackNC State 2025.
Next.js 16, TypeScript, viem, wagmi, Python, Alchemy Account Kit, Sepolia testnet

AlgoPulse – Agentic AI Platform Real-Time + AI

Identified API-polling bottleneck under load; re-architected to event-driven model streaming AI reasoning loops directly to the client via Supabase Edge Functions. Temporal.io and Kafka handle durable state and fault-tolerant worker coordination: 2,000 async requests/second without state loss.
TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

Glance AI – Privacy-First RAG System AI Tooling

Built a compliance-ready RAG system for fintech/healthcare with per-tenant data isolation at the vector retrieval layer, preventing context leakage. Deterministic output architecture ensures factual correctness for regulated data environments.
Python, LLMs, Vector Databases, RAG

PerCent – GenAI Finance App HackNC24 Top 10

Developed a GenAI finance app with goal tracking, expense logging, and a RAG-based chatbot using Qwen-14B-Chat; ranked Top 10 of 100+ teams. Reduced manual financial research time by 30%. Built complete MVP in 48 hours.
Python, Selenium, Hugging Face, Qwen-14B-Chat, RAG

EDUCATION

North Carolina State University – MS Computer Science

Aug 2024 - May 2026

Courses: Artificial Intelligence, Algorithms, Data Science, Operating Systems, Software Engineering, Object-Oriented Development

Dwarkadas J. Sanghvi College of Engineering – BTech Computer Science

Aug 2020 - May 2024

CGPA: 9.32/10 · Courses: Data Warehousing, Big Data Infrastructure, DS & Algorithms, Machine Learning, Software Engineering

SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL, PHP (personal projects), C, R

Frontend: React, Next.js, Tailwind CSS, HTML5, Accessibility fundamentals

Backend & Tools: Node.js, Express, Spring Boot, REST APIs, Docker, GitHub Actions, Jenkins, Git

Cloud & Infra: AWS (EC2, S3, CodeDeploy), GCP (Cloud Run), Grafana, CI/CD, Distributed Systems

Databases: PostgreSQL, MySQL, MongoDB, Redis

AI/ML: OpenAI API, Anthropic Claude, LangChain, MCP, Hugging Face, RAG, TensorFlow, PyTorch